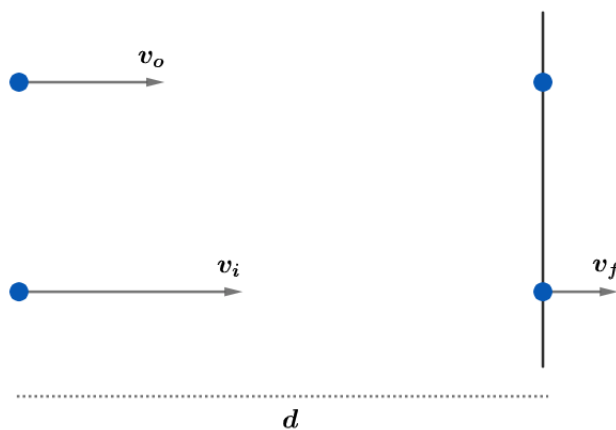


2022A F=ma Exam: Problem 2

Kevin S. Huang



Recall from our kinematics equations,

$$v_f^2 = v_0^2 + 2ax$$

In the first case, we have

$$0 = v_0^2 - 2ad$$

where $v_0 = 60$ mph. In the second case, we have

$$v_f^2 = v_i^2 - 2ad$$

where $v_i = 70$ mph. Substituting the first equation,

$$\begin{aligned} v_f^2 &= v_i^2 - v_0^2 \\ v_f &= \sqrt{v_i^2 - v_0^2} = 36 \text{ mph} \end{aligned}$$

so the answer is D.